

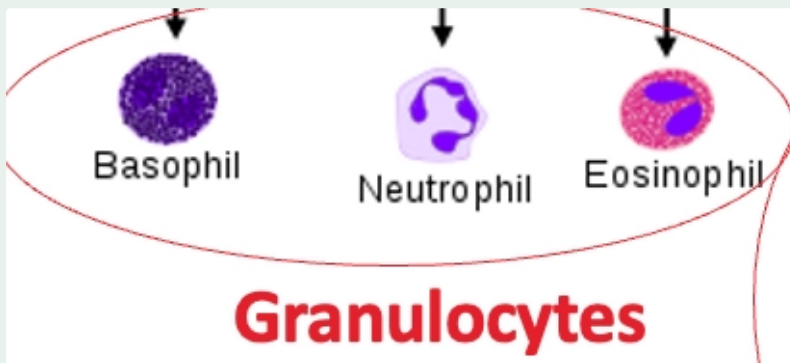
3.5 White Blood Cells

▼ What are granulocytes?

- They are a type of white blood cell that have granules in them.

▼ What are the three main type of granulocyte and their colours?

- **Eosinophil** (red)
- **Neutrophil** (neutral)
- **Basophil** (blue)

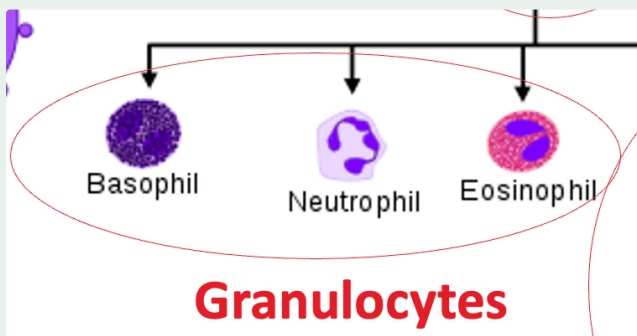


▼ How can we tell that a granulocyte has matured?

- It gets smaller and the development of granules key to its function begins.

▼ Describe the difference in the appearances of eosinophils and basophils.

- Eosinophils are more reddish in colour than basophils which are described as basic (alkaline), giving them a blue colour



▼ What do granulocytes often look like in people with leukaemia?

- They are bigger than usual, immature and they look like **myeloblasts**

▼ Describe the typical differentiation path taken from a HSC to a granulocyte.

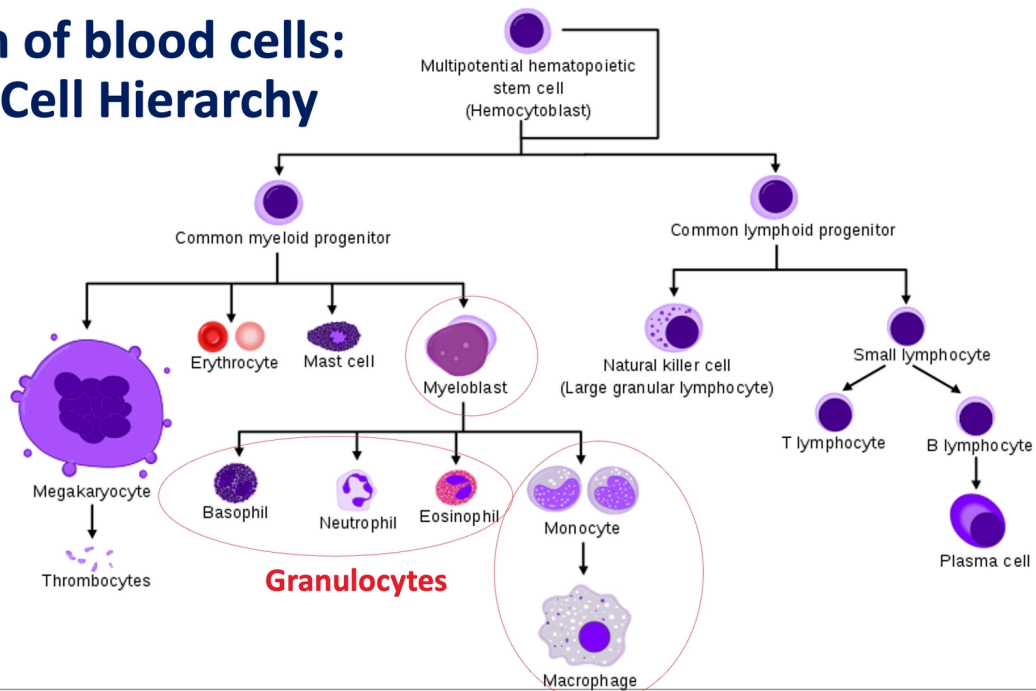
- **HSC → Common Myeloid Progenitor → (Myeloblast) → (Promyelocyte) → (Myelocyte) → (Band form) → Granulocyte**

▼ What is a monocyte?

- They are a type of white blood cell which are used to **destroy pathogens**
- They can **differentiate into macrophages or dendritic cells**.

▼ Diagram showing blood stem cell hierarchy.

Origin of blood cells: Stem Cell Hierarchy



White blood cells: Outline the origin and functions of white blood cells, and recall common causes of abnormalities, including cancer

▼ What can myeloid growth factors be used for?

- They can be used to activate the **proliferation of myeloid (granulocytes + macrophages) cells** and hence are an effective treatment for patients with **pathologies such as leukaemia**.
 - This is only in the case where a leukaemia patient has undergone chemotherapy, and so their lymphocyte count has decreased. Chemotherapy typically kills fast-proliferating cells, so white blood cells can catch collateral damage.
- This is important as these white blood cells are important to the immune system
- **Granulocyte** colony-stimulating factor (G-CSF), **Granulocyte-macrophage** colony-stimulating factor (GM-CSF), **Macrophage** colony-stimulating factor (M-CSF) are examples of growth factors

▼ What is the main role of a neutrophil?

- It is used to **phagocytose (mainly bacteria/fungi)** and then kill these **phagocytosed bacteria/fungi**.

▼ How long does the neutrophil survive for in the bloodstream before migrating to tissues?

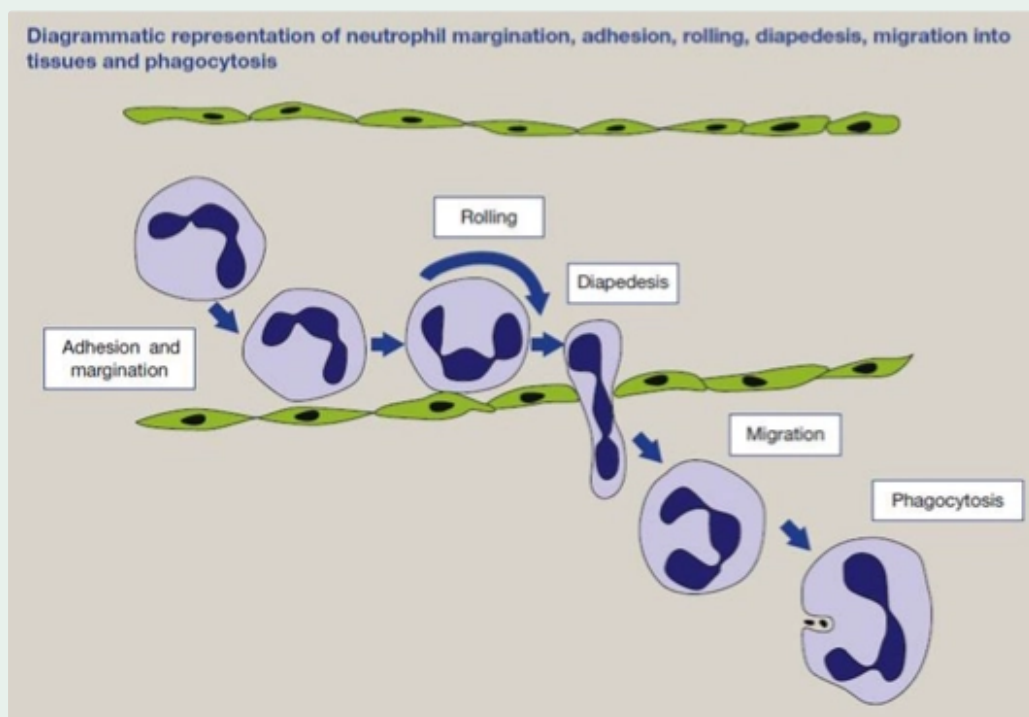
- **7-10 hours.**
- After this period they will migrate to tissues via **chemotaxis**.

▼ What is chemotaxis?

- Chemotaxis is the **movement of an organism or entity in response to a chemical stimulus**.

▼ How does the neutrophil migrate from the bloodstream to the tissues around it, in order to fight infection?

- **Neutrophil extravasation (diapedesis)**
 - Neutrophils, via a complex pathway involving adhesion molecules such as selectins & integrins, cytokines & chemokines and the endothelium, move towards the interstitium and tissue through the endothelial membrane



▼ Which has a longer lifespan, an eosinophil, or a neutrophil?

- Eosinophil.

▼ What are the two main roles of an eosinophil?

- To **defend against parasitic agents** such as helminth worms.
- To **regulate Type I hypersensitivity reactions** (those allergic reactions reacted to in <24 hours)

▼ What is a hypersensitivity reaction?

- They are **exaggerated or inappropriate reactions** conducted by the immune system in response to an **antigen or an allergen**
- It happens when the immune system **mistakes something harmless as something that is harmful**

▼ How do eosinophils regulate these Type I hypersensitivity reactions?

- They **inactivate the histamine and leukotrienes released by basophils** and mast cells.

▼ What is a leukotriene?

- An **inflammatory mediator** released by **basophils**

▼ What is the role of a leukotriene?

- They **contract airway smooth muscle, increase vascular permeability, increase mucus secretion, and attract and activate other inflammatory cells**
- They just cause inflammation really.

▼ What is a special property of basophils?

- Its granules contain **stores of histamine and heparin**, as well as **proteolytic enzymes**.

▼ What are the main roles of a basophil?

- To **act as an inflammatory mediator** and induce inflammatory effects
- To mediate **immediate-type hypersensitivity reactions**
 - i.e Type I

▼ How are basophils involved in mediation of immediate-type hypersensitivity reactions?

- The **Immunoglobulin-E (IgE) coated** basophils **release heparin and proteases**
 - IgE binds to the antigens
- This acts to **mediate immediate-type hypersensitivity reactions**.

▼ How do mast cells differ from basophils?

- **Mast cells** reside in **tissues**
- **Basophils** are **found freely in the circulation**

▼ What is heparin?

- It is an **anticoagulant** that is **naturally occurring** and **can also be administered as a drug**

▼ What does heparin do?

- **Heparin** potentiates the action of **antithrombin** and hence helps to prevent harmful clots from forming in the blood

▼ How do monocytes differ from macrophages?

- **Monocytes** and **macrophages** are two versions of **the same cell**
- A **monocyte** is the name of the **cell when it flows freely in the circulation**
- A **monocyte becomes a macrophage** when it **migrates to the tissue**
 - It is also known as a **histiocyte** in this form

▼ Recall two functions of macrophages.

- **Phagocytosis** of antigenic material
- **Antigen presentation** to T cells. They are **antigen-presenting cells**.
- **Storage and release of iron**

▼ Is it monocytes or macrophages that store iron?

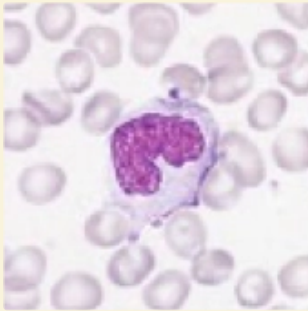
- **Macrophages** (tissue)

▼ What are the two types of cell monocytes can differentiate into after migrating from the blood to the interstitium?

- **Dendritic cells**
- **Macrophages**

▼ What is a characteristic feature of monocytes/macrophages on analysis that allows them to be easily identifiable?

- **Macrophages (not monocytes)** are involved in the **storage and release of iron to erythrocytes**
- Structures called '**erythroid islands**' can **form around** monocytes which allow them to **obtain iron** from these cells.
 - This is the purple "circle" around the main cell body shown below
- You can only see these in the tissues, not the bloodstream, because **macrophages are not found in the tissues**



▼ What type of white blood cell interacts with red blood cells?

- **Macrophages.**

▼ How do macrophages interact with red blood cells?

- They are involved in the **storage and release of iron to erythrocytes**
- Structures called '**erythroid islands**' can form around macrophages which allow them to **obtain iron** from these cells.
 - They **transfer this iron in bone marrow during erythropoiesis**, as **macrophages are not found in the blood** (they would be called monocytes)

▼ What is a plasma cell?

- A **further differentiated B cell** that **makes antibodies** to combat pathogens.

▼ What makes the edge of plasma cells slightly darker?

- The **presence of ribosomes**
- They need this larger number of ribosomes because **they are constantly producing antibodies** (proteins) against pathogenic antigens

▼ Why are lymphocytes found in the lymph nodes?

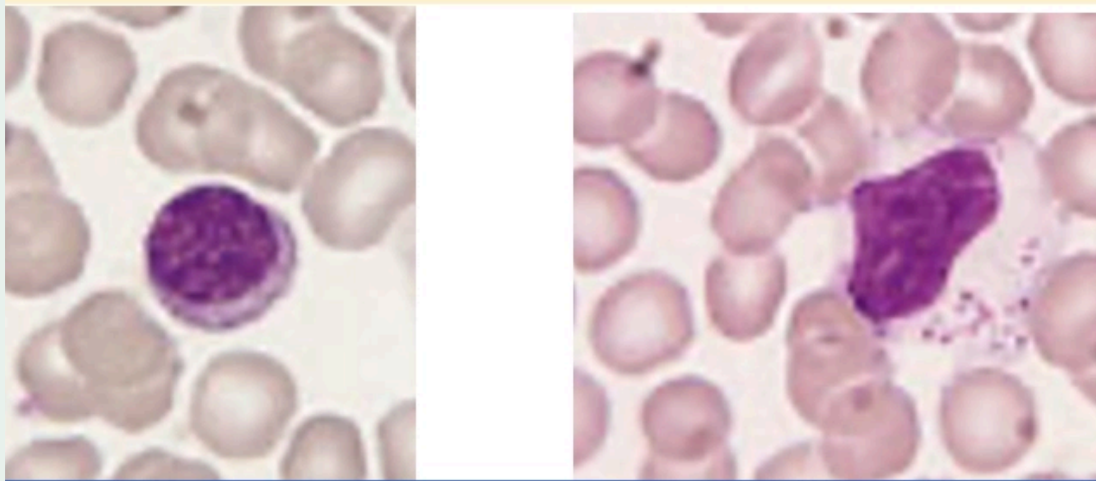
- **Blood is pushed out of the blood vessels** into the **interstitium** and **recollected by lymphatic capillaries**
- The **lymphocytes help to filter blood at the lymph nodes** and this blood is recycled to the circulation

▼ What are the three types of lymphocyte produced by common lymphoid progenitors?

- **Natural killer cells (NK)**
- **T lymphocyte**
- **B lymphocyte**

▼ What is the difference between natural killer (NK) cells and lymphocytes morphologically?

- **Natural killer cells (RIGHT) are bigger than B/T lymphocytes (LEFT)**



▼ Where are B lymphocytes made?

- **Progenitors are made in the foetal liver** before differentiating in the **bone marrow (B)**

▼ What is humoral immunity?

- The immune response that involves the action of **antibodies** which combat different antigens.

▼ How is the development of B lymphocytes centred around the idea of humoral immunity?

- Their development involves the **rearrangement of the genes coding for the heavy and light chains**
- These **different arrangements** lead to **different immunoglobulins** (antibodies) being produced which can fight against different antigens.

▼ How does differentiation work to produce different types of B cells?

- **Differentiation in B cells** involves **Ig heavy and light chain gene rearrangement**
 - This way, the variable region of the antibody is different and a different type of B cell is produced

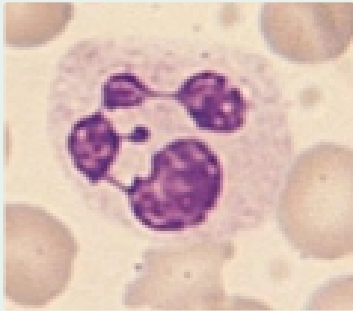
▼ Which type of lymphocytes are involved in cell-mediated immunity?

- **T lymphocytes**
- **Natural killer cells**

▼ Where are T lymphocytes made?

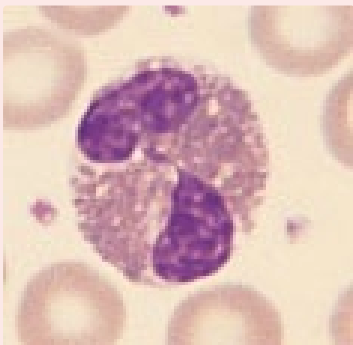
- **Progenitors are made in the foetal liver** before differentiating in the **thymus (T)**

▼ Which blood cell is shown in the image below?



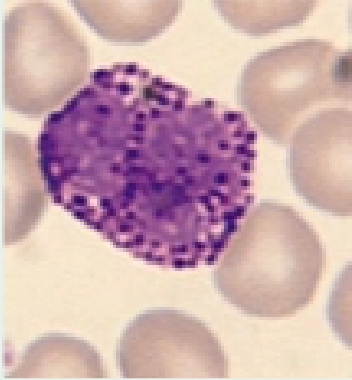
► Answer

▼ Which blood cell is shown in the image below?



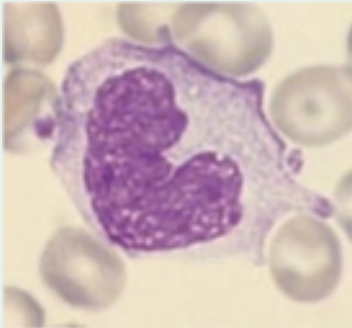
► Answer

▼ Which blood cell is shown in the image below?



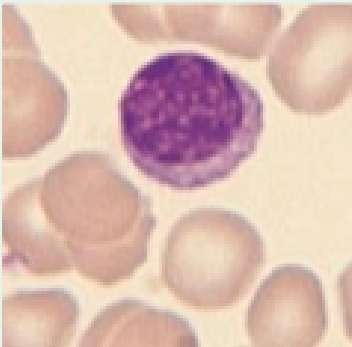
► Answer

▼ Which blood cell is shown in the image below?



► Answer

▼ Which blood cell is shown in the image below?



► Answer

▼ What is leukocytosis?

- The term used to describe an **elevated white blood cell count**.

▼ How can transient leukocytosis help a clinician?

- It can suggest that there is a **secondary (external) cause** for the elevated WBC count
- This could be an **infection**, and the **type of WBC** elevated in number can give us indication of what **type of infection** it is

▼ What type of infection causes neutrophilia/monocytosis ?

- Bacterial

▼ What type of infection causes lymphocytosis?

- Viral

▼ What type of infection causes eosinophilia?

- Parasitic

▼ What does persistent leukocytosis suggest to us?

- There is **damage in the DNA** that affects a precursor lymphoid cell and this is giving rise to blood cancers, **causing elevated WBC levels**.
- **Chronic leukaemias, lymphomas, myelomas**
- It just suggests to us that the **increase in WBC levels is not due to response to pathogens**, but is primary
 - There is no apparent cause for it

▼ What is a myeloma?

- A **cancer of plasma cells (derived from B cells)**

▼ What is leukopenia?

- A reduction in the total number of white cells.
 - It is usually caused by a reduction in neutrophil count (as they are the most prevalent WBC)

▼ What is the most abundant leukocyte in circulation?

- Neutrophils.

▼ Recall three causes of neutrophilia.

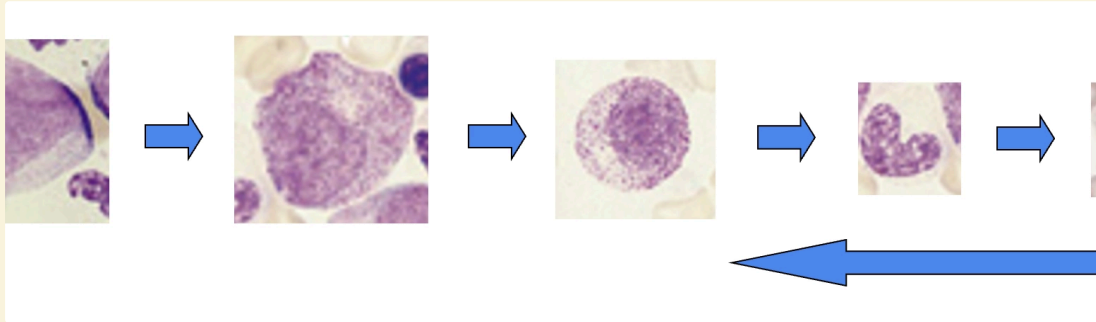
- Infection
- Inflammation
- Tissue damage
- Pregnancy
- Exercise
- Corticosteroids

▼ What adverse effects can neutrophilia lead to?

- **Toxic granulation.**
- **Left shift**

▼ What is a left shift, and what causes it?

- A left shift is an **increase in neutrophil precursors**, usually with **less lobes**, in the blood
 - **Chronic myeloid leukaemias cause left shift. Increased demand on the bone marrow.**



▼ What are the main causes of neutropenia?

- **Radiotherapy and chemotherapy**
- In other cases, it can be caused by **autoimmune disorders** and **severe bacterial infections**.

▼ What can neutropenia lead to?

- Increased **risk of serious infection**
 - These people need **urgent treatment with intravenous antibiotics**

▼ What is neutrophil hypersegmentation and what causes it?

- This is also known as **right shift**
- It is when a neutrophil has **more than the regular 3-5 lobes** (right shift)
- It is caused by a **lack of vitamin B12 or folic acid** which prevents the cell from dividing
 - Similar to B12/folate deficiencies causing macrocytic anaemia in erythrocytes

▼ What can cause eosinophilia?

- Parasitic infection
- Allergy

▼ What can cause basophilia?

- Usually a form of **leukaemia**
 - i.e Chronic myeloid leukaemia (CML)

▼ What is monocytosis?

- **Too many monocytes**

▼ What can cause monocytosis?

- Infection or chronic inflammation.

▼ What causes lymphocytosis?

- Normally, **lymphocytosis** occurs in response to a viral infection

▼ What is the most common lymphocyte type in circulating blood?

- **CD4+ T (helper) cells**

▼ What causes lymphopenia?

- **HIV infection**
- **Chemotherapy**
- **Radiotherapy**
- **Corticosteroids**

▼ What is a mutagen?

- A mutagen is a **chemical or physical agent** capable of **inducing mutations**.

▼ What are the two main types of leukaemia, and how are they classified?

- **Myeloid**
 - i.e granulocytes, monocytes → **Acute/Chronic myeloid leukaemia**
- **Lymphoid**
 - i.e lymphocytes → **Chronic lymphocytic leukaemia/Acute lymphoblastic leukaemia**
- A leukaemia can fall under one of these categories based on **where the mutation leading to the cancer takes place**.

▼ What happens to white blood cells in leukaemia (pathogenesis)?

- The mutant cells have a **distinct survival advantage** over the non-mutant cells, and so **they replace them**
- The **mutant cells proliferate over time**, before taking over the blood and **they are dysfunctional**
- This leads to **patients becoming extremely susceptible to infection**

▼ The four types of leukaemia.

How is leukaemia classified?

The main types of leukaemia are:

- Acute Lymphoblastic Leukaemia (**ALL**, **blast** because immature cells)
- Acute Myeloid Leukaemia (**AML**) (blast cells)
- Chronic Lymphocytic Leukaemia (**CLL**, **cytic** because mature lymphoid cells)
- Chronic Myeloid Leukaemia (**CML**) (mature myeloid cells)

Different types of leukaemia differ in aetiology, nature of the mutational events, age of onset, clinical & haematological features and prognosis

ALL is particularly a disease of childhood – results from somatic mutations *in utero*

CLL predominantly affects the elderly

CML results from the activation of signalling pathways by fusion protein BCR-ABL

White blood cells: Outline the origin and functions of white blood cells, and recall common causes of abnormalities, including cancer

▼ What is the main difference between acute leukaemias and chronic leukaemias?

- **Acute leukaemias** typically occur in **blast (immature) cells**
 - The cells typically **halt due to failure in their life-cycle** and proliferate immaturely
 - Because these immature cells have a higher proliferative capacity, this can lead to propagation of the leukaemia much quicker than in chronic cases
- **Chronic leukaemias** typically occur in **partially mature cells**
 - The cells typically have **partially finished maturing** and proliferate thereafter

- ▼ Diagrams showing the difference between lymphoblastic and lymphocytic leukaemia.

Lymphoid differentiation

